18 An alpha-particle has 2.2×10^{-13} J of kinetic energy.

What is the speed of the alpha-particle?

A $4.1 \times 10^{6}\,\text{m s}^{-1}$

B $5.8 \times 10^{6}\,\text{m s}^{-1}$

C $8.1 \times 10^{6}\,\text{m s}^{-1}$

D $1.2 \times 10^{7}\,\text{m s}^{-1}$